

## SUBNETTING – 5 STEPS

1. Identify the class of the IP address and note the Default Subnet Mask.
2. Convert the Default Subnet Mask into Binary.
3. Note the number of hosts required per subnet and find the Subnet Generator (SG) and octet position.
4. Generate the new subnet mask.
5. Use the SG and generate the network ranges (subnets) in the appropriate octet position.

1, Subnet the ip address 216.21.5.0 into 30 hosts in each subnet

## SOLUTION

1. Class C – Default Subnet Mask: 255.255.255.0
2. 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0
3. No. of hosts/subnet: 30 (11110) – 5 bits SG: 32 Octet Position: 4  
1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 0 0 0 0 0
4. New subnet mask: 255.255.255.224 or /27
5. Network Ranges (Subnets)  
216.21.5.0 – 216.21.5.31  
216.21.5.32 – 216.21.5.63  
216.21.5.64 – 216.21.5.95  
216.21.5.96 – 216.21.5.127  
216.21.5.128 – 216.21.5.159  
and so on....

2, Subnet the ip address 196.10.20.0 into 52 hosts in each subnet

## SOLUTION

1. Class C – Default Subnet Mask: 255.255.255.0
2. 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0
3. No. of hosts/subnet: 52 (110100) – 6 bits SG: 64 Octet Position: 4  
1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 0 0 0 0 0 0
4. New subnet mask: 255.255.255.192 or /26
5. Network Ranges (Subnets)  
196.10.20.0 – 196.10.20.63  
196.10.20.64 – 196.10.20.127  
196.10.20.128 – 196.10.20.191  
196.10.20.192 – 196.10.20.255

3, Subnet the ip address 150.15.0.0 into 500 hosts in each subnet

## SOLUTION

1. Class B – Default Subnet Mask: 255.255.0.0
2. 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0
3. No. of hosts/subnet: 500(11110100)–9 bits SG: 2 Octet Position: 3  
1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 0 . 0 0 0 0 0 0 0 0
4. New subnet mask: 255.255.254.0 or /23
5. Network Ranges (Subnets)  
150.15.0.0 – 150.15.1.255  
150.15.2.0 – 150.15.3.255  
150.15.4.0 – 150.15.5.255  
150.15.6.0 – 150.15.7.255  
150.15.8.0 – 150.15.9.255  
and so on....

$2^9 = 512$  Hosts per Network (Subnet)  
 $2^7 = 128$  Subnets (Networks)